

1. What is Data?

Data are *raw facts and figures* that represent observations, measurements, or descriptions of things. They can be numbers (like height, age) or categories (like gender, colour). Data by itself doesn't mean much until it is processed and analysed to extract meaning or *information*.

2. Major Types of Data

Data is broadly divided into two main types:

1. **Quantitative Data** (Numerical)
2. **Qualitative Data** (Categorical)

A. Quantitative Data (Numerical Data)

This type of data represents *counts or measurements* and answers questions like “**how much?**”, “**how many?**”, “**how often?**”

Two Sub-types of Quantitative Data:

1. Discrete Data

- ✓ *Countable values* usually in whole numbers
- ✓ You *cannot* have fractions within it
- ✓ Counted, not measured

Examples:

- Number of students in a class
- Number of cars in a parking lot
- Number of pets someone owns
(*You can't have 2.5 pets!*)

2. Continuous Data

- ✓ *Measurable values* that can take *any number* (including decimals)
- ✓ Values can lie anywhere within an interval or range
- ✓ Commonly measured, not counted

Examples:

- Height of students (e.g., 170.5 cm)
- Weight of a person (e.g., 65.8 kg)

- Temperature in a room
- Time taken to finish a race

B. Qualitative Data (Categorical Data)

These data represent *qualities or categories* rather than numbers. They describe *attributes or characteristics* and often cannot be used for arithmetic operations.

Types of Qualitative Data:

1. Nominal Data

- ✓ Data that names or *labels categories*
- ✓ *No order or ranking*
- ✓ Categories are *distinct and separate*

Examples:

- Gender (Male, Female, Others)
- Eye colour (Blue, Brown, Green)
- Blood group (A, B, AB, O)
- Types of fruit (Apple, Banana, Mango)

2. Ordinal Data

- ✓ Data that has *order or ranking*
- ✓ You *can compare* categories in terms of sequence
- ✓ But the *difference between categories is not consistent*

Examples:

- Customer satisfaction (Poor, Average, Good, Excellent)
- Educational level (Primary, Secondary, Graduate, Postgraduate)
- Exam grades (A, B, C, D)
- Race positions (1st, 2nd, 3rd)

Key Points to Remember

- ✓ *Quantitative* = numerical data you can measure or count.
- ✓ *Qualitative* = non-numerical data showing categories or qualities.
- ✓ *Discrete* = countable, whole number data.
- ✓ *Continuous* = measurable and can take decimal values.

- ✓ *Nominal* = categories with *no order*.
- ✓ *Ordinal* = categories with *order*, but without consistent spacing.